

Data Structures and Algorithms

Exercise 2-EcommerceSearchExample:

Product.java:

```
class Product {  
    int productId;  
    String productName;  
    String category;  
    Product(int productId, String productName, String category) {  
        this.productId = productId;  
        this.productName = productName;  
        this.category = category;  
    }  
    void display() {  
        System.out.println(productId + " " + productName + " " + category);  
    }  
}
```

SearchDemo.java:

```
class SearchDemo {  
    // Linear Search  
    static Product linearSearch(Product[] products, int id) {  
        for (int i = 0; i < products.length; i++) {  
            if (products[i].productId == id) {  
                return products[i];  
            }  
        }  
        return null;  
    }  
}
```

```
}
```

```
// Binary Search
```

```
static Product binarySearch(Product[] products, int id) {
```

```
    int low = 0;
```

```
    int high = products.length - 1;
```

```
    while (low <= high) {
```

```
        int mid = (low + high) / 2;
```

```
        if (products[mid].productId == id)
```

```
            return products[mid];
```

```
        else if (products[mid].productId < id)
```

```
            low = mid + 1;
```

```
        else
```

```
            high = mid - 1;
```

```
    }
```

```
    return null;
```

```
}
```

```
public static void main(String[] args) {
```

```
    Product[] products = {
```

```
        new Product(101, "Laptop", "Electronics"),
```

```
        new Product(102, "Phone", "Electronics"),
```

```
        new Product(103, "Shoes", "Fashion"),
```

```
        new Product(104, "Book", "Education"),
```

```
        new Product(105, "Watch", "Accessories")
```

```
    };
```

```

int searchId = 104;

System.out.println("Linear Search:");

Product p1 = linearSearch(products, searchId);

if (p1 != null)
    p1.display();
else
    System.out.println("Product not found");

System.out.println("\nBinary Search:");

Product p2 = binarySearch(products, searchId);

if (p2 != null)
    p2.display();
else
    System.out.println("Product not found");
}
}

```

Output:

```

C:\Users\HP\Desktop\Deepskilling\Engineering Concepts\Data Structures and Algorithms\Exercise 2-EcommerceSearchExample>java SearchDemo
Linear Search:
104 Book Education

Binary Search:
104 Book Education

```

Exercise 7-FinancialForecasting:

FinancialForecast.java:

```

import java.util.Scanner;

class FinancialForecast {

    static double futureValue(double amount, double rate, int years) {

        // Base case
    }
}

```

```

    if (years == 0)
        return amount;
    // Recursive case
    return futureValue(amount, rate, years - 1) * (1 + rate);
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter Present Value: ");
    double amount = sc.nextDouble();
    System.out.print("Enter Growth Rate (%): ");
    double rate = sc.nextDouble() / 100;
    System.out.print("Enter Number of Years: ");
    int years = sc.nextInt();
    double result = futureValue(amount, rate, years);
    System.out.printf("Future Value after %d years = %.2f",
        years, result);
    sc.close();
}
}

```

Output:

```
C:\Users\HP\Desktop\Deepskilling\Engineering Concepts\Data Structures and Algorithms\Exercise 7-FinancialForecasting>javac FinancialForecast.java
```

```
C:\Users\HP\Desktop\Deepskilling\Engineering Concepts\Data Structures and Algorithms\Exercise 7-FinancialForecasting>java FinancialForecast
```

```
Enter Present Value: 10000
```

```
Enter Growth Rate (%): 10
```

```
Enter Number of Years: 3
```

```
Future Value after 3 years = 13310.00
```

```
C:\Users\HP\Desktop\Deepskilling\Engineering Concepts\Data Structures and Algorithms\Exercise 7-FinancialForecasting>
```